



Crowd counting with sparse annotation

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ABSTRACT

This paper presents a new annotation method called Sparse Annotation (SA) for crowd counting, which reduces human labeling efforts by sparsely labeling individuals in an image. We argue that sparse labeling can reduce the redundancy of full annotation and capture more diverse information from distant individuals that is not fully captured by Partial Annotation methods. Besides, we propose a point-based Progressive Point Matching network (PPM) to better explore the crowd from the whole image with sparse annotation, which includes a Proposal Matching Network (PMN) and a Performance Restoration Network (PRN). The PMN generates pseudo-point samples using a basic point classifier, while the PRN refines the point classifier with the pseudo points to maximize performance. Our experimental results show that PPM outperforms previous semi-supervised crowd counting methods with the same amount of annotation and achieves competitive performance with fully-supervised methods.

1. Introduction

Crowd counting [1–3] is a crucial task with significant applications in public security and transportation systems. Recent advances in Neural Networks (CNNs) [4] have improved the performance of crowd counting methods. However, most of them [5–7] are fully supervised and require complete image annotations, making the annotation process challenging and time-consuming. Crowd counting annotation requires marking every head in an image with a point. This process becomes even more cumbersome when dealing with images with dense populations, where heads are small and difficult to distinguish. For example, annotating the UCF-QNRF dataset [8], which contains 1535 images with an average of 815 heads per image, requires over 2000 h.

Semi-supervised crowd counting [9–11] methods, as shown in Fig. 1(a), can release the burden of annotation, in which a small part of the training images is fully annotated. However, it cannot capture the diversity on the image level because there still exist many images in the unlabeled set that are different from the labeled images. To better utilize the information on every image, partially annotated crowd counting [12] employs a patch of full annotations to train the networks as Fig. 1(b) shows. Although it incorporates more diverse information among training images with the same number of labels, it overlooks the diversity of different regions on each image. In fact, the difference between individuals in a patch is significantly less than those from distant individuals. According to the imaging principle

of “close large, far small”, distant individuals may have completely different scales. So partial annotation that can only label persons in one patch overlooks the diversity on the patch level. Therefore, we propose a new annotation setting for crowd counting in this paper named **Sparsely Annotated Crowd Counting**. It not only captures the diversity on the image level by sparsely annotating every image but also incorporates the diversity at the patch level by annotating every region in an image. As shown in Fig. 1(c), we randomly and sparsely label the individuals across the whole image instead of fully annotating in a patch. Furthermore, we propose two kinds of sparse annotation. One is a *ratio*-based sparse annotation which uses a pre-defined *ratio* to control the number of annotated head points. The other one is *K*-points based sparse annotation, which takes a fixed number *K* as the maximum number of annotated head points. Compared to prior methods that either do not fully utilize all training images, or they do not adequately preserve intra-image diversity across regions and scales, our method utilizes every training image and preserve the intra-image diversity.

Most crowd counting methods are density map based [1–3]. A Gaussian kernel on every head dot is used to generate the density map. However, density map based methods are counter-intuitive and error-prone [5]. In practice, an accurate density map cannot be generated due to the annotation being sparse in our setting. To deal

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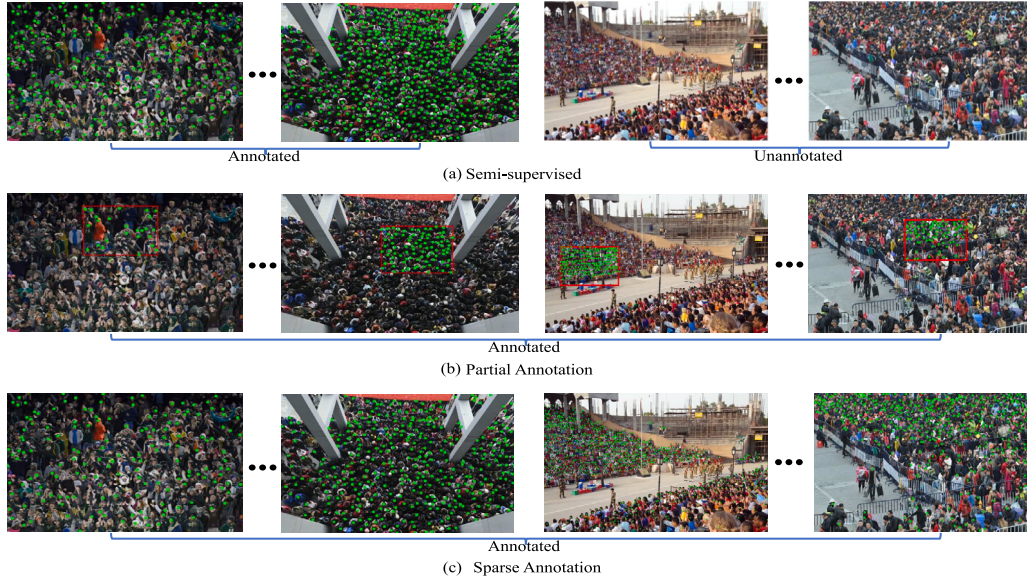


Fig. 1. Three settings for crowd counting with limited annotation are depicted from top to bottom as follows: (a) Traditional **semi-supervised setting** involves annotating all individuals for some images and leaving the rest images unlabeled. (b) **Partial Annotation** using a fully annotated patch for each image. (c) The proposed **sparse Annotation** involves random annotating individuals across the image.

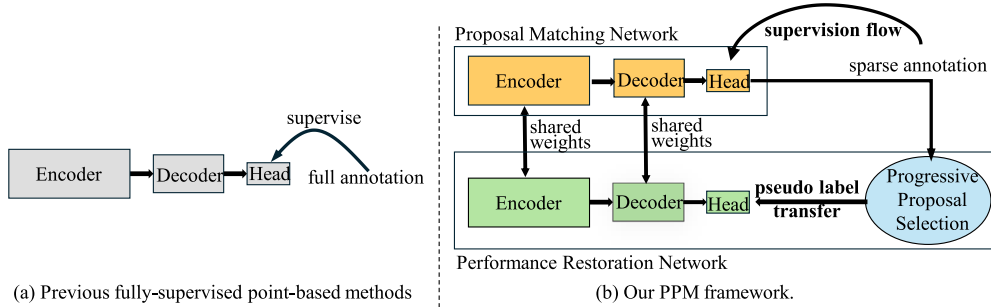


Fig. 2. Comparison with the pipeline of previous fully supervised point-based methods.

with this, we developed a point-based framework named Progressive Point Matching Network (PPM). It consists of a Proposal Matching Network (PMN) and a Performance Restoration Network (PRN). In the PMN, a point classifier is trained to predict a confidence score for each proposal. One-to-one matching is leveraged to assign the point proposals to ground truth. The performance of PMN is significantly influenced when the annotated ratio is low. The fewer labels we have, the easier the classifier is affected by noisy labels. To exclude the inaccurate labels and explore the benefits of the sparse labels, we incorporate a progressive proposal selection (PPS) strategy to choose pseudo samples during the training procedure. Hence, the PRN, a distillation classifier that learns from confident pseudo samples with high scores is proposed. Meanwhile, a confidence-guided weighted loss is applied to adaptively assign weight to the positive proposal to further alleviate the impact of noisy point samples. As Fig. 2 demonstrates, compared with previous fully supervised point-based methods that only have one branch, our pipeline following a dual-branch scheme which is specially designed for sparse annotation. We conduct extensive experiments to evaluate the efficiency of the proposed PPM approach under the sparse annotation setting on five challenging datasets of ShanghaiTech_A [1], ShanghaiTech_B [1], UCF_QNRF [8], JHU-Crowd++ [13] and NWPU-Crowd [14] and two point-based baselines. Several ablation and robustness experiments are

also conducted to make comprehensive evaluation. The results show that our method outperforms all of the previous semi-supervised crowd counting methods by a large margin. Moreover, it only needs 80% annotations to achieve competitive performance with previous state-of-the-art fully-supervised methods and 90% annotations to exceed them. Therefore, beyond proposing a method, we introduce a meaningful supervision setting that provides a clear trade-off between annotation cost and model performance.

In a nutshell, our main contributions can be summarized as follows:

1. We propose a novel crowd counting annotation setting called **Sparsely Annotated Crowd Counting** and two kinds of sparse annotation to reduce the annotation cost and capture the diversity of image level and patch level.
2. We design a simple yet effective framework named Progressive Point Matching network (PPM), consisting of a Proposal Matching Network to train a basic point classifier supervised by sparse annotation and a Performance Restoration Network to refine the point classifier.
3. The proposed PPM approach on Sparsely Annotated Crowd Counting setting achieves competitive performance with the state-of-the-art fully-supervised crowd counting methods and outperforms all the semi-supervised methods by a large margin.

2. Related work

2.1. Fully-supervised crowd counting

In recent years, based on the rapid development of Deep Convolutional Neural Network, fully-supervised crowd counting [2,5,15] gains satisfying performance.

Density map based method was first introduced in [16], where they used all of the point labels to produce a pseudo density map to supervise the predicted density map of the network. The estimated number is obtained by summing over the predicted density map. Recently, a large amount of studies have focused on density map schemes, most of them solving various challenges in crowd counting and pushing forward the counting performance frontiers. [1] proposed a multi-column model to regress the predicted density map, and designed an adaptive Gaussian kernel to generate pseudo density maps. To deal with extremely congested scenes, [3] fused dilated kernels into crowd counting networks to estimate the count. [17] introduced an attention network to crowd counting to extract attentive features. In [2], the Bayesian assumption was introduced and Bayesian loss was proposed to estimate the count expectation at each annotated point. In [18], optimal transport was proposed to match the distributions. [19] designed a trainable confusion region discriminating and erasing network to learn the difference of feature representations in confusion regions. [20] designs a super-resolution enhancement network to alleviate the impact of low resolution on counting performance.

Except for density map based methods, some new solutions were also proposed. [5] discarded density maps and proposed a purely point-based network, using Hungarian Matching to make a one-to-one matching. In addition, [21] proposed Mean Count Proxies to select the best count proxy for each interval to represent its count value during inference. [6] introduces transformer into crowd localization, proposing a crowd localization transformer to directly predict the localization. [22] leverages auxiliary point guidance to overcome the instability in training. [15] views crowd counting as a decomposable point querying process.

These fully supervised methods achieve incredible performance on various datasets. However, the complete annotation for a crowd image is time-consuming and laborious. In UCF-QNRF dataset [8], it costs over 2000 human hours to annotate only 1535 images, in which the average cost per image is over one hour.

2.2. Weakly/semi-/un-supervised crowd counting

In order to relieve the crowd counting annotation efforts, an increasing number of weakly/semi-/un-supervised crowd counting methods have been proposed recently.

In [23], it collected many unlabeled crowd images from the internet and leveraged them to train a learning-to-rank framework to estimate the density maps for crowd counting. Furthermore, [24] focused on learning to count from a limited number of annotated samples, and proposed a Gaussian Process-based iterative learning mechanism. [9] tackled the problem from the perspective of feature learning, unlabeled images are used to train a generic feature extractor. In [10], the spatial uncertainty-aware teacher-student framework was proposed to deal with the noisy supervision from unlabeled data. In [11], a density agency and a contrastive learning loss were proposed to recognize foreground features and consolidate feature extractor. [25] developed a multi-task credible pseudo-label learning framework for crowd counting. Moreover, [12] decided to label one patch of an image rather than the entire image. Besides, [26] propose a prototype-based learning approach with an embedded understanding of perspective. [27,28] leverages count-level labels as weak supervisions. [29] formulates the pixel-wise density value to regress as a probability distribution.

In this paper, we propose a solution that is significantly distinct from all of the above methods. Specifically, to capture the diversity on both the image level and patch level, Sparsely Annotated

Crowd Counting(SACC) is designed. With random and sparse annotation, we propose a Progressive Point Matching network (PPM) to explore the benefits of sparse annotation. The solution outperforms all previous semi-supervised methods with the same amount of annotation and obtains competitive performance to fully-supervised methods with fewer annotations which achieve a balance between performance and annotation cost.

3. The setting of sparse annotation

We denote the training image set for crowd counting as $\mathcal{X}$ with the head point $\mathcal{P}$ and test image set $\mathcal{Y}$ with counting number $\mathcal{N}$. Given an image $I \in \mathcal{X}$ with N people, the set of head point is $\mathcal{P} = \{p_i\}_{i=1}^N \in \mathcal{P}$, where $p_i = (x_i, y_i)$ is the coordinate of the center point for the i th person.

Traditional semi-supervised methods completely label part of the images in the training set, but they ignore the image level difference. Partial annotation setting increases the diversity on the image level by employing a patch of full annotation. However, it overlooks the difference in the patch level. By observing that the difference between individuals in a patch is largely less than those from distant individuals, we propose a novel setting of sparse annotation to capture the diversity on both the image level and the patch level. As shown on the right of Fig. 1, in this new setting, we randomly and sparsely annotate the individuals across the whole image.

Then, we propose two kinds of sparse annotation, one is *ratio* based sparse annotation(*r*-annotation) and the other one is *K* points based sparse annotation(*K*-annotation).

***r*-annotation.** As the accurate N cannot be obtained for the training image in our setting during annotation, we can roughly estimate the total number $\hat{N}^1$ and ask the annotator to label $N^c = \text{ratio} \times \hat{N}$ people.² The *ratio* controls the number of head points to be annotated, which is steerable to make a trade-off between performance and annotation cost.

***K*-annotation.** *K* points based sparse annotation is a more straightforward annotation method. For *K*-annotation, we set a fixed number *K* as the upper limit number of annotating head points for each image. If the total number of an image is less than *K*, we will annotate all of the heads in the image. Hence, we do not need to estimate the total number.

For *r*-annotation and *K*-annotation, the set of annotated head points is denoted as $\mathcal{P}^c \in \mathcal{P}^c$ and $\mathcal{P}^c$ is a subset of $\mathcal{P}$. The ultimate goal is to train a model on $\mathcal{X} \times \mathcal{P}^c$ to predict a counting number on $\mathcal{Y} \in \mathcal{Y}$ which is close enough to the accurate counting number $N \in \mathcal{N}$.

4. The proposed method

4.1. Overview

To deal with sparsely annotated crowd counting, we utilize a point-based scheme rather than generating inaccurate density maps due to the nature of sparse annotation. We propose a simple yet effective Progressive Point Matching network (PPM), as shown in Fig. 3. The training image is passed through a VGG-16 Network [30] to produce feature maps. Then these features are fed into a Proposal Matching Network(PMN) to train a basic point classifier generating pseudo head point samples. To exclude the inaccurate labels and explore the benefits

¹ In practical scenarios, such a rough estimate can be easily obtained through a quick visual scan when the total count is relatively small. And when the total number is large, we can use a simple strategy to obtain the rough count (e.g. we can calculate the ratio by estimating the area covered by the annotated points relative to the total area of all persons.)

² In this paper, we directly random select $N^c = \text{ratio} \times N$ points as sparse annotation from the completely annotated dataset. A simple random sampling strategy is applied.

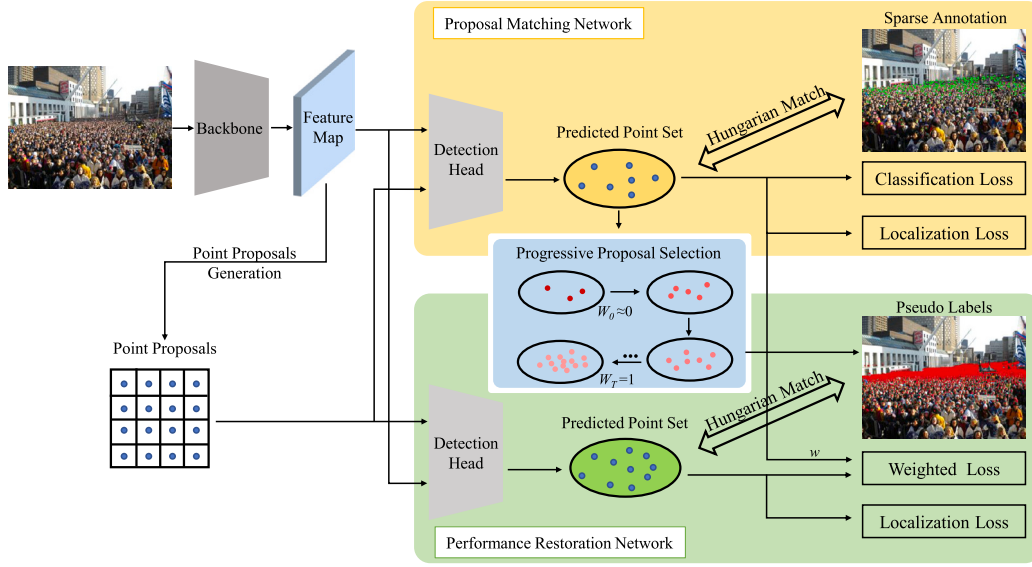


Fig. 3. The overall architecture of the Progressive Point Matching network. Feature map extracted by the backbone is used to generate point proposals. The features and proposals are input into two parallel networks. The Proposal Matching Network trains a basic point classifier that produces a coarse prediction and provides pseudo point samples. The Performance Restoration Network gets a distillation classifier, which is supervised by the pseudo points. A Progressive Proposal Selection strategy and a confidence-guided weighted loss are used to alleviate the adverse impact of noise in pseudo samples.

of the sparse labels, we incorporate a Progressive Proposal Selection strategy to pick out the pseudo samples. At last, we propose a Performance Restoration Network (PRN) as a distillation classifier to learn from the confident pseudo samples with high scores.

In the inference stage of PPM, the test image is forwarded into the PRN to predict the head points. The total number is generated by counting the predicted heads.

4.2. Proposal matching network

The PMN is used to produce a coarse prediction by the supervision of sparse annotation $\mathcal{P}^c$ and provides pseudo point samples for the PRN, as shown in Fig. 3.

Point Proposal Generation. Supposing the down-sampling rate of the backbone is s , each location of the feature map is corresponding to a $s \times s$ patch of the input image. To densely detect the head points, Z point anchors are used at the center of the patch. The point proposals are generated by sliding the point anchors on the feature map. We denote the set of point proposals as $R = \{r_j | j \in \{1, \dots, M\}\}$, where r_j is a point proposal with location (x_j, y_j) and M is the total number of point proposals.

Point Detection. Similar to object detection, point detection is accomplished with two parallel branches for point classification and point coordinate regression. The classification branch predicts the confidence scores $\hat{c}_j$ for every point proposal. The regression branch outputs M pairs of location offsets $(\Delta x_j, \Delta y_j)$. The set of predicted points is denoted as $\hat{P} = \{\hat{p}_j | j \in \{1, \dots, M\}\}$, where $\hat{p}_j = (x_j + \Delta x_j, y_j + \Delta y_j)$. With the confidence score and point coordinates, a one-to-one matching is conducted between the predicted points $\hat{P}$ and sparse annotation $\mathcal{P}^c$ with the Hungarian algorithm [31] of $\Omega(\mathcal{P}^c, \hat{P}, D)$. The pair-wise matching cost matrix D is defined as $D(\mathcal{P}^c, \hat{P}) = (v \| p_i - \hat{p}_j \|_2 - \hat{c}_j)$, where $p_i \in \mathcal{P}^c$ and $\hat{p}_j \in \hat{P}$, $\| \cdot \|_2$ is Euclidean distance, and v is a weight term. With the one-to-one matching, the i th annotated point is assigned to one predicted point $\hat{p}_{\sigma(i)}$, where $\sigma(i)$ is the index of the matched point. The unmatched are labeled as negatives. The total loss of Proposal Matching Network is defined as:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{loc}, \quad (1)$$

where λ_1 is a weight factor to balance the two losses, following P2PNet [5]. The classification loss $\mathcal{L}_{cls}$ is a Cross Entropy Loss defined as:

$$\mathcal{L}_{cls} = -\frac{1}{M} \left(\sum_{i=1}^{N^c} \log \hat{c}_{\sigma(i)} + \lambda_2 \sum_{N^c+1}^M \log(1 - \hat{c}_{\sigma(i)}) \right), \quad (2)$$

where λ_2 is a weight factor to balance the negative proposals. The location regression loss $\mathcal{L}_{loc}$ is Euclidean distance loss defined as:

$$\mathcal{L}_{loc} = \frac{1}{N^c} \sum_{i=1}^{N^c} \| p_i - \hat{p}_{\sigma(i)} \|_2^2. \quad (3)$$

4.3. Performance restoration network

PMN provides a basic point classifier and regressor, but the performance is greatly affected when the annotated ratio decreases. To exclude the inaccurate labels and explore the benefits of the sparse annotation, we incorporate a progressive proposal selection strategy to choose pseudo samples during the training procedure, as shown in Fig. 3. Meanwhile, the PRN, a distillation classifier that learns from the confident pseudo samples with high scores is also proposed to refine the point classifier. The PMN and PRN are trained end-to-end. It consumes less training time than that in an alternative way.

Hard Threshold for Pseudo Points. The predicted points of the PMN are noisy as the number of proposals M is much larger than the number of annotated points N^c . Thus, it is critical to choose reliable pseudo points. Intuitively, we first propose a straightforward method to choose the positive point proposals with a hard threshold. For a point $\hat{p}_j$, the confidence score vector $\hat{c}_j$ is a 2D vector that represents the confidence of belonging to the background and foreground respectively. The hard threshold τ is used on the confidence score of the foreground to select pseudo points. For a point $\hat{p}_j = (x_j, y_j)$, its label is defined as:

$$L_{pseudo} = \begin{cases} 1, & \text{if } \hat{c}_j > \tau \\ 0, & \text{otherwise} \end{cases}. \quad (4)$$

The pseudo point samples are selected by using this simple hard threshold on all of M predicted points, which provides supervision for the PRN.

Progressive Proposal Selection. Hard threshold can largely improve the performance of the baseline, however, because the score for

each point proposal is changing during training, it is better to change the threshold value along with the increasing of training epochs. Hence we propose a Progressive Proposal Selection(PPS) strategy to exclude inaccurate labels and explore the benefits of the sparse labels. Denote r as a point sample with confidence score c , $0 \leq W_t \leq 1$ as the weight applied to r at epoch t with $0 \leq t \leq T$ satisfying $W_0 \approx 0$ and $W_T = 1$. With the increase of W_t , the distribution of training samples becomes more complex for the PRN. The pseudo points are selected by a progressive threshold depending on W_t . The label of the point is defined as:

$$L_{pseudo} = \begin{cases} 1, & \text{if } c \geq \tau_1 - \tau_2 \cdot W_t \\ 0, & \text{otherwise} \end{cases}, \quad (5)$$

where τ_1 and τ_2 are constant values satisfying $0 \leq \tau_2 < \tau_1 \leq 1$ to limit the progressive threshold to a section $[\tau_1 - \tau_2, \tau_1]$.

Weighted Loss. The loss functions of the PRN is similar to the PMN as Eq. (1). A Hungarian Matching is conducted between the pseudo points and the point proposals so that the matched proposals are labeled as positive and the others are negative. The number of pseudo-point labels is denoted as N^s . The loss of location regression is updated as:

$$\mathcal{L}_{loc} = \frac{1}{N^s} \sum_{i=1}^{N^s} \|p_i - \hat{p}_{\delta(i)}\|_2^2. \quad (6)$$

A weighted classification loss is used in the PRN, defined as:

$$\mathcal{L}_{cls} = -\frac{1}{M} \left(\sum_{i=1}^{N^s} w \log \hat{c}_{\delta(i)} + \lambda_2 \sum_{i=1}^M \log(1 - \hat{c}_{\delta(i)}) \right), \quad (7)$$

where w denotes the scores of proposals outputted by the PMN. When the learning begins, the pseudo labels are not accurate and a small weight w is applied to alleviate the issue of over-fitting to noise. With the training epochs increasing, the prediction is more accurate and reliable, and the higher weight w is applied. It brings the advantage that the positive ones are easier to recognize accurately.

5. Experiment

We first introduce the datasets and the experimental setting including implementation details, and evaluation metrics. Then, the performance is compared with fully-supervised, semi-supervised and partially annotated crowd counting approaches. At last, the effectiveness of the proposed Sparse Annotation and counting approach is discussed in the ablation study.

5.1. Datasets

Our experiments are conducted on five publicly available datasets.

ShanghaiTech_A(SHA) [1] consists of 482 images with 244,167 annotated persons. 300 images are used for training and the rest of 182 images are used for testing. The images of SHA are randomly collected from the internet. SHA is used for the ablation study in this paper.

ShanghaiTech_B(SHB) [1] includes 716 images with 88,498 annotated persons which means its crowd density is lower than SHA. 400 images are used as the training set and the remaining 316 images are used as the test set. All of the images are collected in the crowded streets of Shanghai.

UCF_QNRF(QNRF) [8] is composed of 1535 high resolution images with 1.25 million annotated persons. 1201 images are used for training and 334 images are used for testing. Since the average number of persons per image is 815 and the maximum number is 12,865, QNRF is a crowd dataset with high crowd density. Furthermore, because of the diverse scenes, perspectives, crowd densities, and illumination conditions it has, QNRF is a very challenging dataset.

JHU-Crowd++(JHU) [13] is a large-scale dataset that contains 4372 images with 1.51 million annotations. There are 2272 images used for training, 500 images for validation, and the remaining 1600

images are used for testing. The images are collected under different conditions and locations. Moreover, there exist some images with weather changes and illumination changes, which makes JHU a very difficult dataset.

NWPU-Crowd(NWPU) [14] is a large-scale and challenging datasets for crowd counting. It contains 5109 images and 2.13 million annotations. There are 3109 images used for training, 500 images are used for validation, and 1500 images are used for testing. In the dataset, the number of annotated people in a single image ranges from 0 to 20,033, which means it has intense scale variations. And the images of NWPU are collected under different conditions which makes it have diverse scenes. Furthermore, 351 negative samples of nobody or textures similar to congested crowd scenes are included in the dataset to add some new challenges to the dataset.

5.2. Experimental setting

Implementation details. We implemented our method on two point-based approaches, P2PNet [5] and PET [15]. For P2PNet, the number of anchor points Z is set as 8 for the QNRF dataset and 4 for the others. Following [5], the weight term v is set as 5e-2, λ_1 and λ_2 are set as 0.05 and 0.4. The training epochs is set as 2000. We use Adam optimizer to optimize the network with a fixed learning rate 2e-4. For NWPU dataset, model that has best MAE on validation set is selected. For other datasets, model of the last epoch is selected. And a fixed threshold 0.5 is adopted during inference. For PET, the learning rate is 10e-4, and the initial stride of sparse point queries is set to 8. For the two baselines, the batch size is 8. For the hard threshold, τ is set to 0.5. For PPS, $W_t = e^{(\theta t - \beta)}$. t indicates t th epoch. θ is a scale factor while β is a bias term. θ is set as 0.05 and β is 10. T is 200. τ_1 and τ_2 are set as 0.6 and 0.2 empirically. The random seed is fixed, the same sparse subsets are reused across experiments, and the reported results on SHA are averaged over 3 runs.

We initially adopt random scaling which is selected from [0.7, 1.3]. Then we randomly crop patches from a resized image. For SHA, SHB, JHU and NWPU, the patch size is set to 128×128 . For QNRF, the patch size is 256×256 . In the end, a random horizontal flipping with a probability of 0.5 is adopted. For QNRF and NWPU, we keep the length of the longer side smaller than 1408 and 2048 respectively to keep the original aspect ratio.

Evaluation metrics. We follow the commonly used metrics in crowd counting, using Mean Absolute Error (MAE) and Mean Squared Error (MSE) for evaluation. MAE and MSE can be formulated as follows:

$$MAE = \frac{1}{N_a} \sum_{i=1}^{N_a} |C_i - GT_i| \quad (8)$$

$$MSE = \sqrt{\frac{1}{N_a} \sum_{i=1}^{N_a} |C_i - GT_i|^2} \quad (9)$$

where N_a is the number of testing images. C_i is the predicted number of the i th image. And GT_i is the ground truth count of i th image.

5.3. Performance comparison

Comparison with Semi-supervised Methods. The setting of crowd counting with Sparse Annotation is a special case of semi-supervised manner. The typical semi-supervised setting is with complete annotation on part of the training images, while ours is with Sparse Annotation on each image of the training data.

Firstly, we compare our method with the state-of-the-art semi-supervised methods on SHA dataset with 40%, 10% and 5% annotations, as shown in Table 1. The results are averaged over 3 random samplings which means the sparse subset is different in each training runs. The results show that we obtain the best or second best performance. Our results surpass TMTB [36] with 40% annotations and

Table 1

Comparison with the semi-supervised methods on SHA dataset. Our results are averaged over 3 random samplings.

Method	Venue	40%		10%		5%	
		MAE	MSE	MAE	MSE	MAE	MSE
PA [12]	ICCV2021	70.5	105.0	72.8	111.6	79.4	123.6
DACount [11]	MM2022	68.2	116	82.5	123.2	85.4	134.5
OT-M [32]	CVPR2023	70.7	114.5	80.1	118.5	83.7	133.3
CU [33]	ICCV2023	64.7	109.6	70.8	116.6	74.5	127.5
HPS [34]	TIP2024	64.7	106.3	73.9	123.5	–	–
SALCrowd [35]	MM2024	<u>60.7</u>	<u>97.2</u>	69.7	114.5	–	–
P ³ Net [29]	TPAMI2025	63.0	100.9	72.1	116.4	85.5	131.0
PPSC [26]	PR2025	62.6	96.6	68.9	107.1	72.7	107.7
TMTB [36]	CVPR2025	60.8	99.0	65.7	<u>110.4</u>	72.4	126.4
P2PNet+PPM(ours)	2025	63.8(±0.2)	111.4(±0.5)	78.6(±0.3)	123.7(±0.6)	81.5(±0.2)	132.1(±0.4)
PET+PPM(ours)	2025	60.4(±0.1)	97.9(±0.5)	<u>68.0(±0.2)</u>	111.9(±0.4)	<u>72.6(±0.1)</u>	<u>115.6(±0.5)</u>

Table 2

Comparison with semi-supervised crowd counting methods on QNRF dataset.

Method	40%		10%		5%	
	MAE	MSE	MAE	MSE	MAE	MSE
DACount [11]	91.1	153.4	109.0	187.2	120.2	209.3
OT-M [32]	100.6	167.6	113.1	186.7	118.4	195.4
SALCrowd [35]	95.0	167.0	106.7	<u>171.3</u>	–	–
P ³ Net [29]	90.0	155.4	<u>103.4</u>	179.0	<u>115.3</u>	<u>195.2</u>
P2PNet+PPM(ours)	95.3	168.4	107.8	174.3	116.2	196.6
PET+PPM(ours)	<u>91.1</u>	<u>159.7</u>	100.5	169.6	112.7	191.2

Table 3

Comparison with fully-supervised (FS) approaches. With 80% sparse annotations (SA), our method achieves comparable performance to the two baseline methods: P2PNet [5] and PET [15]. With 90%, the performances are further improved consistently on all datasets and is comparable to the best method APGCC [22]. The best scores are in bold and the second ones are with an underline. The results are obtained by averaging over three runs, with standard deviations shown in parentheses.

Method	Type	Ratio	SHA		SHB		QNRF		JHU++	
			MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
CSRNet [3]	FS	100%	68.2	115.0	10.6	10.6	108.2	181.3	85.9	309.2
BL [2]	FS	100%	62.8	101.8	7.7	12.7	88.7	154.8	75.0	299.9
DMCount [18]	FS	100%	59.7	95.7	7.4	11.8	85.6	148.3	68.4	283.3
GL [37]	FS	100%	61.3	95.4	7.3	11.7	84.3	147.5	59.9	259.5
ChfL [38]	FS	100%	57.5	94.3	6.9	11.0	<u>80.3</u>	<u>137.6</u>	57.0	235.7
P2PNet [5]	FS	100%	52.7	85.1	6.5	10.5	85.3	154.5	–	–
PET [15]	FS	100%	<u>49.3</u>	78.7	6.1	9.6	79.5	144.3	58.5	238.0
APGCC [22]	FS	100%	48.8	76.7	5.6	8.7	80.1	136.6	54.3	225.9
P2PNet+PPM(ours)	SA	90%	51.4(±0.2)	84.7(±0.7)	6.4(±0.1)	10.7(±0.2)	85.0(±0.3)	144.3(±0.9)	56.5(±0.2)	233.1(±1.3)
P2PNet+PPM(ours)	SA	80%	52.3(±0.3)	85.4(±0.9)	6.6(±0.1)	10.8(±0.1)	85.4(±0.3)	146.9(±1.3)	58.8(±0.2)	234.5(±1.5)
PET+PPM(ours)	SA	90%	49.3(±0.3)	78.6(±0.5)	6.0(±0.2)	9.3(±0.3)	80.6(±0.4)	145.5(±1.5)	58.4(±0.3)	238.7(±1.6)
PET+PPM(ours)	SA	80%	50.5(±0.5)	80.8(±1.1)	6.5(±0.3)	10.2(±0.4)	81.2(±0.4)	147.5(±1.3)	59.5(±0.4)	240.8(±2.1)

Table 4

Comparison with fully-supervised approaches on the NWPU-Crowd.

Method	Ratio	Val		Test	
		MAE	MSE	MAE	MSE
CSRNet [3]	100%	104.9	433.5	121.3	387.8
BL [2]	100%	93.6	470.4	105.4	454.2
DMCount [18]	100%	70.5	357.6	88.4	388.6
P2PNet [5]	100%	–	–	77.4	362.0
PET [15]	100%	–	–	74.4	328.5
P2PNet+PPM(ours)	90%	63.9	311.1	76.2	350.4
P2PNet+PPM(ours)	80%	65.1	311.9	76.8	351.6
PET+PPM(ours)	90%	62.2	307.3	74.2	326.3
PET+PPM(ours)	80%	64.8	308.7	75.1	334.9

are slightly higher than it with 10% and 5% annotations, since TMTB adopts VMamba [39], a much more powerful visual state space model as its backbone while our backbone is VGG16.

Furthermore, we compare with semi-supervised methods on the large-scale dataset QNRF in Table 2. With lower annotation ratios, such as 10% and 5%, our proposed method achieves state-of-the-art

performance which got 2.9 and 2.6 points improvement on MAE. The experimental results verify the superiority of our method.

Partial Annotation (PA) [12] is a crowd counting annotation setting that chooses to annotate one patch for each image in the training set. It is also a special case of semi-supervised manner. In Table 1, our method is also compared to it. The results show that our

Table 5
Comparison of the localization performance on NWPU-Crowd.

Method	Ratio	Val Pre	Rec	F1-m	Test Pre	Rec	F1-m
CLTR [6]	100%	73.9	71.3	72.6	69.4	67.6	68.5
FIDTM [40]	100%	<u>73.7</u>	76.7	<u>70.9</u>	<u>70.5</u>	74.4	<u>66.9</u>
Ours	90%	75.7	<u>72.4</u>	74.0	71.8	<u>70.1</u>	70.9
Ours	80%	73.2	70.7	71.9	69.8	67.9	68.8

Table 6
Comparison of the localization performance on SHA.

Method	Ratio	$\sigma = 4$ Pre	Rec	F1-m	$\sigma = 8$ Pre	Rec	F1-m
CLTR [6]	100%	43.6	42.7	43.2	74.9	73.5	74.2
FIDTM [40]	100%	59.1	58.2	58.6	<u>78.2</u>	<u>77.0</u>	<u>77.6</u>
Ours	90%	<u>56.2</u>	<u>56.3</u>	<u>56.2</u>	79.4	79.5	79.5
Ours	80%	54.4	53.3	53.8	79.7	77.9	78.8

method gains 10.0 and 6.7 points improvement with 40% annotations on MAE and MSE, respectively. And Table 1 shows that we receive 4.6 and points improvement on MAE with 10% and 5% annotations, respectively. Therefore, quantitative results indicate that our sparse annotation, which captures diversity on both the image and patch levels, outperforms partial annotation with the same annotation ratio.

Comparison with Fully-Supervised Methods. We extensively compare our method with the fully-supervised methods on SHA, SHB, JHU and QNRF, as shown in Table 3. The results are obtained by averaging over three runs, the sparse subset in each run is the same. With only 80% annotations, we consistently reach the fully supervised baselines on all datasets. If the annotation ratio is 90%, the MAE and MSE are further improved, and we achieve competitive results with the state-of-the-art method.

The results on the large-scale NWPU [14] dataset are illustrated in Table 4. On the validation set, the proposed method achieves best performance with an MAE of 62.2 and an MSE of 307.3 with 90% annotations on the validation set. On the test set, we also achieved competitive performance with an MAE of 74.2 and an MSE of 326.3. These demonstrate that our method is robust in various crowd counting scenes.

The satisfying performances even compared to fully-supervised methods verify the effectiveness of our sparse annotation.

5.4. Analysis

In this section, we conduct experiments on the SHA dataset with P2PNet [5] to show the performance of crowd localization and cross-dataset generalization, the performance across different ratios, the effect of PMN, PRN, PPS and Weighted Loss, the impact of annotation ratio disturbance.

Performance of crowd localization. As aforementioned, our method is also capable of predicting crowd localization based on our PPM framework. In this section, we make a comparison with other methods of crowd localization.

As shown in Table 5, we achieve the best crowd localization performance on NWPU dataset. With 90% annotations, we achieve the best performance with 75.7 Precision, 72.4 Recall, 74.0 F1-measure on validation set and 71.8 Precision, 70.1 Recall, 70.9 F1-measure on test set. While with an annotation ratio of 80%, the performance is also better than CLTR [6].

Then, we make a comparison between our method and other Point Matching crowd localization methods on SHA dataset. As Table 6 shows, we also obtain the best performance with $\sigma = 8$ and the second best performance with $\sigma = 4$.

Performance of cross-dataset generalization. To evaluate the cross-dataset generalization ability, we compare our approach with baselines on the SHA and SHB [1] datasets. As Table 7 demonstrates,

with 40% annotations, our method surpasses DACount and SALCrowd by 2.7 and 1.1 MAE when adapting SHA to SHB. And we outperform DACount and SALCrowd by 6.4 and 2.6 MAE when adapting SHB to SHA.

The performance across different ratios. We conduct experiments on different annotation ratios and the corresponding K to verify the effectiveness of our method. To be fairer, we adjust the value of K to ensure that the total annotation number of r -annotation and K -annotation are approximately equal. The results are shown in Table 8 and we visualize the results in Fig. 4. With complete annotation, the baseline miraculously achieves an MAE of 52.7 and MSE of 85.1, which is an impressive performance of fully supervised crowd counting. However, the MAE and MSE drastically deteriorate along with the decrease of the annotation ratio, especially when the ratio is lower than 50%.

The reduction of annotation makes the baseline under-fitting. In sharp contrast, our methods trained with r -annotation on different ratios (red curve) show a big advantage over the baseline. When the annotation ratio is 90%, the MAE and MSE are miraculously better than the fully-supervised baseline. Following the decrement in the annotation ratio, the MAE and MSE of our approach increased slightly from 51.3 to 72.2 and from 84.3 to 134.7, respectively. Besides, 80% annotations are enough for our method to surpass the performance of the fully-supervised baseline, which saves 20% annotation labor. The proposed setting of sparse annotation and approach is steerable to achieve a trade-off between the crowd counting performance and the annotation labor.

However, our methods trained with K -annotation on different K shows a decline over the baseline when the *ratio* is bigger than 70%. We think this shows that r -annotation is better than K -annotation and the unbalance between different scales of K -annotation is severe when K takes a large value. Although K -annotation shows a decline when the K is large, it is better than the baseline when the *ratio* is small. When the annotation ratio is 60%, K -annotation receives an advantage of 13.2 MAE and 19.6 MSE over the baseline, which shows the effectiveness of our framework.

The comparison between r -annotation and K -annotation shows that our methods trained with r -annotation obtains a better performance. Hence, r -annotation is used as our annotation method in the following experiments.

The effect of PMN and PRN. In Table 9, we study the effect of Proposal Matching Network (PMN) and Performance Restoration Network (PRN). As shown in Table 9, compared to the baseline, both the performance of PMN and PRN are improved. This indicates that PMN and PRN promote each other during training.

The effectiveness of PPS and Weighted Loss. As shown in Table 10, the experiments study the effectiveness of the Progressive

Table 7

The evaluation of cross-dataset generalization.

Method	SHA→SHB	MSE	SHB→SHA	MSE
	MAE		MAE	
DACount [11]	128.4	131.0	203.7	331.9
SALCrowd [35]	126.8	129.3	199.9	328.1
Ours	125.7	127.9	197.3	325.6

Table 8

The performance of our approach across different annotation ratios or K values. The value of K is adjusted to ensure an equal total annotation number. The baseline is trained with r -annotation. The MAE and MSE of the baseline are drastically degraded along with the decrease of the annotation ratio, while the performance degradation of our approach is slight.

Ratio	Baseline		r -annotation		K -annotation	
	MAE	MSE	MAE	MSE	MAE	MSE
100%	52.7	85.1	—	—	—	—
90%($K = 1300$)	53.8	90.5	51.3	84.3	57.8	93.8
80%($K = 800$)	56.3	95.1	52.2	84.9	59.9	101.2
70%($K = 600$)	65.1	112.4	54.5	91.7	66.7	126.1
60%($K = 450$)	85.7	153.7	57.3	94.6	72.5	134.1
50%($K = 330$)	120.2	207.9	59.5	103.9	80.6	143.8
40%($K = 250$)	165.5	252.2	63.9	111.2	91.7	157.1
30%($K = 180$)	228.4	316.1	67.5	121.2	101.3	175.6
20%($K = 120$)	336.3	438.4	72.2	134.7	112.6	198.3

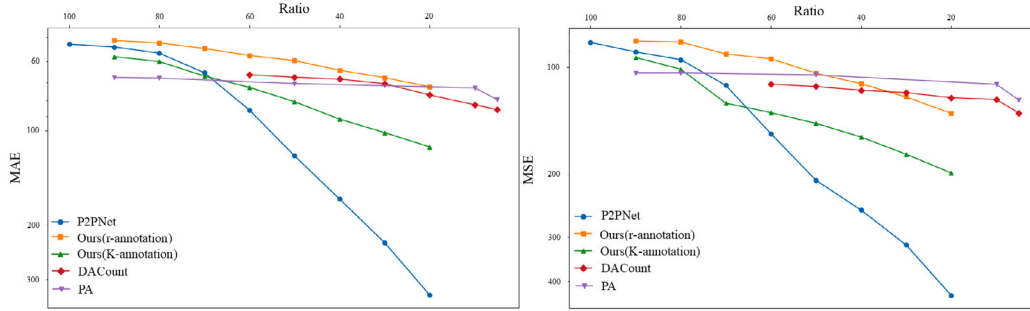


Fig. 4. The MAE and MSE with different ratio of annotation for baseline P2PNet [5], semi-supervised DACount [11], partial annotation [12], r -annotation and K -annotation. It illustrates that our proposed method is closer to the upper bound of fully supervised baseline.

Table 9

The ablation of PMN and PRN on SHA.

Ratio	Baseline		PMN(ours)		PRN(ours)	
	MAE	MSE	MAE	MSE	MAE	MSE
80%	56.3	95.1	53.3	86.9	52.2	84.9
70%	65.1	112.4	55.7	93.6	54.5	91.7

Table 10

Ablation study of Weighted Loss and PPS.

	Weighted loss	MAE	MSE
Baseline w. 60%		85.7	153.7
Hard Threshold		59.9	103.7
Hard Threshold	✓	58.3	101.3
PPS		58.4	98.5
PPS	✓	57.3	94.6

Proposal Selection (PPS) strategy and Weighted Loss for the Performance Restoration Network. In these experiments, the annotation ratio is set to a fixed value of 60%.

A simple hard threshold $\tau = 0.5$ largely improves the baseline method with 60% annotations. The MAE decreases from 85.7 to 59.9 and the MSE decreases from 153.7 to 103.7. These verify that the Performance Restoration Network is efficient. As the score for each proposal changes during training, we utilize the Progressive Proposal Selection strategy, in which an exponential function is used to control the change rate of parameter W_r . The MAE and MSE are reduced by 1.5

points and 5.2 points, respectively. When the weighted loss is used, the MAE and MSE are further reduced by 1.6 points and 2.4 points, respectively. The MAE and MSE have further decreased to 57.3 and 94.6 with 1.71% and 3.95% relative improvements when both PPS and Weighted Loss are adopted.

Impact of annotation ratio disturbance. When the annotator is asked to label the head points in an image with an accurate ratio for a crowd counting dataset, there must be some disturbance so that an approximate ratio of heads is annotated. We simulate this disturbance by applying different Gaussian distributions and using the 3σ rule

Table 11
The impact of annotation disturbance on SHA.

Disturbance	Ratio range	80%		60%	
		MAE	MSE	MAE	MSE
$N(0, 3)$	± 0	52.2	84.9	57.3	94.6
$N(0, 5)$	$\pm 5\%$	52.6	86.5	57.7	95.3
$N(0, 11)$	$\pm 7\%$	52.7	86.7	58.3	96.7
$N(0, 11)$	$\pm 10\%$	52.8	88.1	58.6	99.7
$N(0, 25)$	$\pm 15\%$	53.0	89.9	58.7	99.2

Table 12
The number of annotated persons under different annotation setting.

	90%	80%	50%
Partial Annotation	154K	140K	97K
Sparse Annotation(ours)	146K	130K	81K

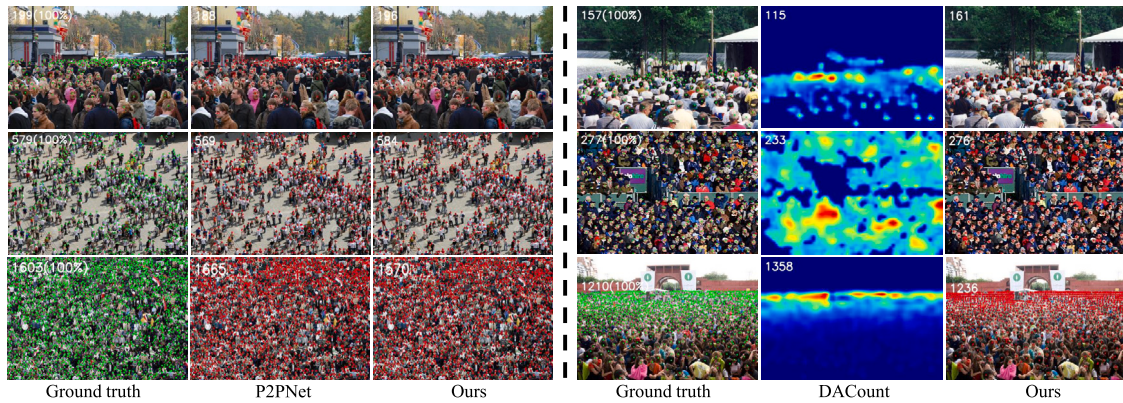


Fig. 5. Visualization of crowd counting prediction. On the left, it illustrates that our method is better than the baseline. On the right, it shows our method achieves better estimation than the semi-supervised DACount [11]. As our method is point-based, it also gives the head location of people.



Fig. 6. Visualizations of failure cases. All are produced under 5% annotation ratio.

to vary the range with different σ values. For example, when the disturbance follows the $N(0, 11)$ distribution, the required annotation ratio is 80% while the actual ratio ranges from 70% to 90%.

From Table 11, we can conclude that the disturbance of the annotation ratio does not have a significant impact on the performance of our model. With 80% annotations, the MAE increases by about 0.4 points for the range $\pm 3\%$, decreases by 0.5 points for the range $\pm 5\%$, and decreases 0.6 points and 0.8 points respectively for the range $\pm 10\%$ and $\pm 15\%$. With a 60% annotation ratio, the largest increase observed is 1.4 points, which is relatively small. These demonstrate that the proposed approach is robust when there exists a disturbance of the annotation ratio. Consequently, when a new crowd counting dataset is annotated under our sparsely annotated setting, an expected accurate annotation ratio is not necessary and an approximate one is sufficient.

The number of different setting. We count the number of annotated persons of partial annotation and sparse annotation in Table 12. The results indicate that the annotation costs of sparse annotation is relatively lower.

5.5. Visualization

On the left of Fig. 5, we visualize the results of our model and the baseline. The baseline is improved significantly as we use the Performance Restoration Network to refine the point localization model. On the right of Fig. 5, it shows the visualization results of our method and DACount [11] with semi-supervision. Our method achieves a better estimation of the count than DACount. Besides, different from the density map-based semi-supervised method which only estimates the number, our method also predicts the head localization of people as we take a point-based manner.

In Fig. 6, we show some failure cases in the proposed method. All of these cases are produced under 5% annotations because our method under very low annotation ratio is inclined to miss the detections rather than false positives. And improving the performance under extremely low annotation ratio can be a future direction for sparse annotation crowd counting.

6. Conclusion

Crowd counting is suffering from burdensome data annotation, which limits its application in real scenarios. To release the cost of annotation, we propose a novel Sparsely Annotated Crowd Counting setting which explores the impact on the performance of the sparse annotations in each image. Moreover, two different sparse annotation settings are proposed. We also propose a simple yet effective framework to accomplish this setting, which consists of a Proposal Matching Network and a Performance Restoration Network. A basic point classifier is trained in the Proposal Matching Network, which is used to generate pseudo point samples. Performance Restoration Network refines the classifier with the pseudo points. The extensive experiments on public crowd counting datasets verify the effectiveness of our setting and approach. The sparse annotation is limited to crowd counting in this paper, and it could potentially extend to other point-supervision tasks such as cell counting in histopathology or object counting but requires careful redesign. And the performance at low ratios still need further improvement to approach the fully-supervised setting.

CRediT authorship contribution statement

Shiwei Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Zhengzheng Wang:** Visualization, Software. **Qing Liu:** Writing – review & editing. **Fei Wang:** Writing – review & editing. **Wei Ke:** Writing – review & editing, Supervision, Funding acquisition. **Tong Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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